Proxy-Based Deep Learning Framework for Spectral–Spatial Hyperspectral Image Classification: Efficient and Robust

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***Abstract*— Deep convolutional networks have been extensively deployed in hyperspectral image (HSI) classification. Reaching for high accuracy, the existing deep-learning-based methods commonly deepen or widen their networks for better performance, which brings higher computational complexity and the risk of overfitting. Although the introduction of the residual module and batch-normalization reduces the generalization degradation in complex networks, the mainstream methods still suffer from low robustness to the noise. To tackle these issues, a compact proxy-based deep learning framework is proposed to perform highly accurate HSI classification with superb efficiency and robustness. In this article: 1) novel deep proxies are integrated to replace the dense classifier layers in conventional networks, which represents specific classes in deep embedding space and enables fast and reliable convergence; 2) the proxy-based feature embedding is studied in distance metric and similarity metric, and compatible dual-metric loss functions are designed for further optimized embedding distribution, which leads to more robust generalization; and 3) state-of-the-art performance and robustness are demonstrated by the proposed framework on mainstream HSI data sets with the minimal network scale and time complexity.**

***Index Terms*— 3-D deep learning, convolutional neural networks (CNNs), hyperspectral image (HSI), HSI classification, proxy-based learning.**

I. Introduction

DUE to the outstanding spectral information density [1]–[5], hyperspectral image (HSI) data have attracted major attention from the remote sensing research community in past decades, and applications based on HSI data have covered the gambit of disciplines in Earth observations from agriculture [6], [7], water [8], and mineral analysis [9] to surveillance services [10]. Among the implementation of these applications, HSI classification is an indispensable technology. The mission of HSI classification is to distinguish a certain land-cover type for each pixel in the HSI data according to the spectral characteristics and its spatial relations.

Even though the abundant spectral bands enable the HSI classification to perceive subtle differences in the spectrum, the high dimension of HSI data acquires superior capability in key feature extraction, which brings obstacles for expanding HSI application. Apart from the high spectral redundancy, several external factors also bring unavoidable difficulties to the HSI analysis. The primary challenge is the noise in HSI data. Hyperspectral sensors usually capture HSI images passively. The spectrum that enters the camera is composed of the radiation of reflected sunlight. During the process of capturing the intake light, the noises caused by the camera distortion and atmospheric activity are inevitable. Another challenge is the nonuniform pixels (outliers). Remote sensing HSI data are mostly captured on airborne or spaceborne platforms, and the shooting range and the limited spatial resolution bring difficulties to distinguish the subpixel ground instances, which leads to the nonuniformity in HSI pixels. These nonuniform pixels increase the proportion of outliers.

In the last few decades, massive studies have been done to tackle the problems in HSI classification. Generally, there are two types of HSI classification methods based on the different ways of feature extraction: handcrafted and deep learning. Most of handcrafted feature-based methods used conventional classifiers, such as support vector machine (SVM) [15], random forest (RF) [16], and extreme learning machines (ELMs) [17]. To efficiently represent the high-dimensional data, methods such as low-rank representation [18]–[20], sparse representation [21], and kernel method [22] were also got applied.

In recent years, the convolutional neural network (CNN) has become the state-of-the-art method for the strong fitting and representation ability [23], as well as the end-to-end learning architecture. After several years of being developed, the mainstream HSI classification methods are able to use fewer training samples to exceed the accuracy of traditional methods more. Some recent works have pushed the state-of-the-art accuracy to a fairly high level. For example, HybridSN [11]

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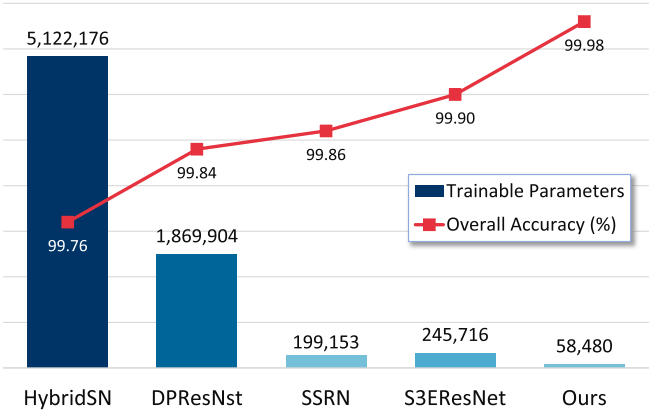


Fig. 1. Overall accuracy on the University of Pavia data set and the scale of parameters of the proposed method and several state-of-the-art methods: HybirdSN [11], SSRN [12], DPResNet [13], and S3EResNet [14]. PCA is integrated in HybridSN, S3EResNet, and our framework for dimension reduction before feature extraction. Our proposed proxy-based deep learning framework achieves the highest performance while significantly reducing the network parameters demanding.

achieves less than 1% overall error rate on all mainstream data sets by using 15% of training data. Most recently, graph convolutional network (GCN)-based approaches [24]–[26] are also emerged rapidly, aiming at thorough exploration of the long-range relations between the HSI samples. Though the deep learning methods have gained huge success in HSI classification, there are still several drawbacks. One of them is low efficiency. To reach satisfactory performance, several works tend to integrate a deeper or denser convolutional network, which increases the network parameters and the difficulty to converge. For example, SSRN [12] uses a ResNet-8-based network with eight layers of 3-D convolution, including nearly 200k trainable parameters in total, and it needs hours to train. Another problem is overfitting caused by the complex network; mainstream CNN implicitly learns the representation of data through layers of the network, which causes high sensitivity to the outliers and noises. By introducing deep learning techniques, such as residual block [12]–[14], [27], dropout [14], [28], and batch-normalization (BN) [14], the ability of network on generalization and antioverfitting has been noticeably improved. However, when facing the data that have been significantly affected, the balance between performance and efficiency is still a serious problem for deep learning-based HSI classification methods.

To tackle the obstacles above, a deep proxy-based classification framework is proposed in this article. At the front end of the proposed framework, a small-scale 3-D convolutional network is incorporated to extract the spectral–spatial semantic feature from the HSI data cube. At the back end, each class is represented by the proxy, which is a vector representing a certain category of training data and being trained as the last layer of the network. In our framework, the classification is simply achieved by aligning the embedded sample feature with the closet proxy. Moreover, based on different measurements, we also propose a series of loss functions that are designed to guide and optimize the 3-D convolutional network and proxies to the ideal state efficiently and robustly. Compared to the existing CNN-based HSI classification methods, even though the HSI data are heavily interfered, our framework is still able to reach high accuracy with less computational resource consumption. As shown in Fig. 1, the proposed method outperforms several state-of-the-art methods with minimal network scale when using 15% training samples and 11×11 input patch on the University of Pavia data set. Overall, there are several contributions worth being highlighted in this article.

1) A compact proxy-based deep learning framework is proposed to achieve efficient and robust spatial–spectral HSI classification. We utilize trainable deep proxies to substitute the dense classification layers of CNN, which is a more straightforward way to represent the class in embedded feature space and brings superb convergence and robustness to the network.
2) Multiple proxy classification loss functions are proposed to train the network and proxies jointly. Based on the insights of proxy-based feature embedding on two major metrics, these proxy loss functions are designed for compact intraclass distribution and clear interclass boundary, which is crucial for precise and robust proxy-based learning.
3) In empirical evaluation, by using a simple four-layer 3-D CNN, the proposed framework reaches comparable or even better performance with the state-of-the-art CNN-based HSI classification methods and also shows great advantages in terms of time complexity, parameter efficiency, and robustness.

The rest of this article is arranged as follows. In Section II, mainstream deep learning HSI classification methods and several proxy-based learning methods are reviewed. Then, the details of the proposed framework and matching loss functions are introduced in Section III, and the performance of the framework is evaluated and discussed in Section IV. In the end, some conclusions about this work are drawn in Section V.

II. Related Works

A. Deep Learning-Based HSI Classification

Deep learning models have been extensively introduced to the remote sensing community, demonstrating superiority in classification-related tasks. The main idea behind deep learning is to extract deep discriminative feature by stacked neuron layers, which enables the model with the understanding of high-level semantic information. A survey and thorough evaluations about deep learning approaches for HSI classification can be found in the literature [2].

For HSI classification, extensive deep learning approaches have been explored, such as stacked autoencoders (SAEs) [29], deep belief networks (DBNs) [30], recurrent neural networks (RNNs) [31], and CNN [32]. Specifically, researchers [33], [34] utilize a series of autoencoders to extract deep spectral–spatial feature in an unsupervised manner. DBN uses the deep Boltzmann machine (DBM) [35] to enhance the discriminative ability in high-dimensional input while requiring dimension reduction at the front end. For HSI classification

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DBN, Chen *et al.* [36] employed principal component analysis (PCA) as the front end, while Arsa *et al.* [37] integrated another DBN to perform the dimension reduction. RNN [31] is originally designed for sequential data, while, for HSI, the high spectral dimensions provide sequence-like property. RNN is applied to HSI classification, for the first time, by Mou *et al.* [38]. Subsequently, Hang *et al.* [39] proposed a cascaded RNN model combined with gated recurrent units, which is capable to learn from the spectral domain and the spatial domain simultaneously.

Nowadays, most state-of-the-art HSI classification methods are integrating CNN-based architecture [5], [23]. The first spectral–spatial CNN-based approach is accomplished by Yue *et al.* [40], which is a hybrid of PCA, CNN, and logistic regression (LR). For the 3-D cube data structure of HSI data, 3-D CNN is widely studied for the advantage in spectral–spatial feature extraction. The first attempt in this direction is achieved by Chen *et al.* [41], where a 3-D CNN is combined with L2 regularization, resulting in a noticeable network complexity reduction. Later in the literature [28], multiple 3-D CNN setting schemes are designed and evaluated, which further proves the effectiveness of 3-D convolution. Recently, HybridSN [11] reaches state-of-the-art level performance by a sequential 3-D–2-D CNN, where 3-D CNN facilitates the joint spatial–spectral feature representation at the early stage, 2-D CNN is integrated for more abstract-level spatial representation. To prevent the overfitting phenomenon from the scale expansion the network structure, active learning [42], residual-block [12], [13], [27], and their variants [43] are introduced to improve the robustness in HSI classification. Lately, CapsuleNet [1] incorporates the capsule block to model the correlation inside the HSI data, which effectively reduces the complexity of the network designing, and also reaches top-level performance.

Though previous deep learning methods have gained remarkable progress, the proposed method still enhances the CNN-based HSI classifier in three ways, which are better computing/parameter efficiency, higher accuracy, and extraordinary robustness.

B. Proxy-Based Deep Learning

Proxy-based learning or prototype learning is one of the most classical and intuitional machine learning methods. It emerges from k-nearest-neighbor (KNN) [44], one of the most famous proxy-based learning methods. While KNN is groundbreaking and successful, it requires massive memory space to restore the parameters, and the optimization process is too complicated to apply. To simplify KNN, learning vector quantization (LVQ) [45] is proposed to reduce the consumption of memory and computation. The literature [44] is a comprehensive review of conventional proxy-based learning algorithms. Previous proxy-based learning methods were mostly developed before the upsurge of CNN and using the handcrafted feature in extraction. To explore the potential of proxy-based learning, several studies have combined this classical method with CNN and the probabilistic model. Yang *et al.* [46] proposed a convolutional network enhanced proxy-based learning, which classifies the instances by the similarity between the input to learned proxies. Bonilla and Robles-Kelly [47] model the input as a mixture of probabilities and deploy a probabilistic model to parameterize proxies by likelihood maximization. The combination of CNN and proxy also can be found in recent few-shot learning studies [48], [49] while showing great robustness when the training data are extremely limited. Recently, in metric learning [50], proxy-based learning is becoming a mainstream solution for the low computational complexity. In the work proposed by Movshovitz-Attias [51], the introduction of proxies significantly reduces the training time, and the resulting models achieve new state-of-the-art level performance.

Despite the proxy-based CNN gains advantages on efficiency and robustness, the overall performance of the proxy-based network heavily relies on the appropriate metric and loss estimation [48]. In our work, two major metrics are explored in HSI classification, i.e., the Euclidean distance and cosine similarity, and multiple loss functions are designed for high-representative proxy distribution.

III. Methodology

Generally, reaching high HSI classification performance with a shallow CNN model is challenging. For HSI classification, in addition to the ultrahigh dimension in the spectral dimension, noise and outliers are inevitable. Mainstream CNN-based methods usually integrate deep and dense convolution architecture to enhance the discriminant ability of the network, thereby expanding the network scale and increasing the risk of overfitting, which will greatly affect performance [46]. This issue can be mitigated using techniques such as residual modules [27], BN, and dropout [52]. Unfortunately, even coupled with the abovementioned modules, when facing highly noisy data, existing CNN-based classifiers still suffers from the high diversity of embedding feature (as shown in Table IX), thus degrading the robustness in generalization. To address these problems, a deep proxy-based learning framework is designed for robust and accurate HSI classification while maintaining efficiency in the architecture and computational complexity.

A. Overview

The overview of the proxy-based deep learning framework is shown in Fig. 2. The whole framework consists of spectral–spatial feature extraction module, the proxy learning module, and multiple loss functions matched with the proxy learning. The data flow starts from a spatial–spectral patch sampler, which samples 3-D patches out of the dimension-reduced HSI data cube. Then, the 3-D convolutional feature extractor encodes each patch into an embedded feature vector. Then, a series of proxies are established in this embedding feature space by the mean value of specific early training samples, and each of them represents a ground-cover class of HSI. After initialization, the proxy learning module is trained by the backpropagation of the loss functions based on the differences among sample feature the proxies. To thoroughly explore the potential in the deep proxy-based HSI

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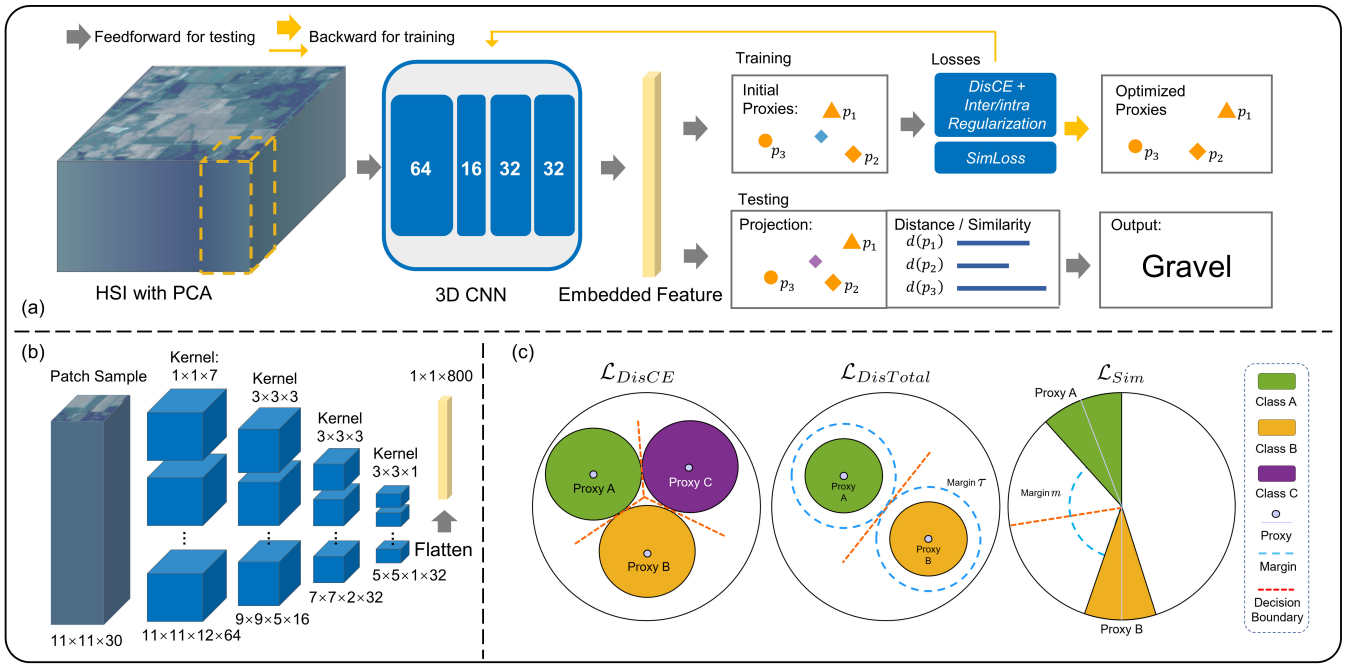


Fig. 2. Overview of the proposed approach. (a) Proxy-based deep learning framework for HSI classification. $p1$–$p3$ represent three proxies, respectively. It can be seen that after the optimization is completed, the proxy of the corresponding category is closer to the labeled sample, and the proxies of different categories will be pushed away. During the training, only single-metric loss functions are applied at a time. (b) Detailed network layers' setting. The scale at bottom of the picture is the output feature map size of each layer. (c) Illustration of the proxy-based feature space distributions when optimized by DisCE loss, DisTotal loss, and Sim loss correspondingly.

classifier, two major metrics are studied in the perspective of the proxy geometric distribution. Based on the characteristics of different metrics, a series of loss functions are designed for intraclass compactness and interclass expansion in training.

In the following part of this section, we will describe the proposed framework from three perspectives. The preprocessing and feature extraction are described in Section III-B, and the workflow and strategy of proxy-based deep learning are introduced in Section III-C. For two types of loss metrics that we designed for the framework, they are explained adequately in Section III-D.

B. Spectral–Spatial Feature Extraction

To reach the high efficiency for convolutional feature extraction, one of the most straightforward ways is to reduce the trainable parameters of the network. For convolutional networks, the parameter compactness of the network is positively correlated with the wideness (amount of channels of each layer), deepness (the number of layers), and the kernel size of convolution. Following this, our proposed framework tries multiple ways to simplify the convolutional computing to a dainty architecture, which results in the most efficient deep learning HSI classification methods as far as we have known.

1) Preprocessing: For shrinking the channels from the very beginning of the networks, the HSI raw data $\mathbf{H}^{w\times h\times d}$ are dimension reduced and decorrelated by PCA before fed into the convolutional network. The dimension reduction is performed on the spectral domain to reduce the spectral redundancy, which results in a c channels of data cube $\mathbf{D}^{w\times h\times c}$. The applied decorrelation can be viewed as a coordinate transformation to the principal component direction, which makes the distribution of data more conducive to the subsequent procedure, while maintaining the integrity of the data. For the categories with very limited samples, minimal information loss may result in significant differences in performance. After dimension reduction, the HSI data cube is sampled by the decorrelated data into partial overlapping patches. To jointly use spatial–spectral information, the data sampler takes each labeled pixel associated with the $n \times n$ neighborhood and all c channels, forming a data patch set $\{\mathbf{F}_m^{n\times n\times c}, m \in 0, 1, \ldots, N_{\text{labeled}}\}$.

2) 3-D CNN Architecture: In the early stage of deep-learning-based HSI classification, so as other image-based pattern recognition tasks, mainstream feature extraction methods usually employ stacked 2-D convolution kernels. However, the 2-D-kernel-based convolutional layer lacks the power of comprehensive analyzing the 3-D form data [53]–[55], such as RGB-D, video, and multispectral image. To enhance the feature extraction ability of CNN, 3-D convolution kernel has demonstrated the power of extracting effective spatial–spectral features of HSI data in multiple works [11], [12], [56].

To extract distinguishing characteristics of input samples, following the mainstream approaches, a multilayer 3-D convolution network is adapted as the feature extractor of the framework. In order to adapt to data and scenes of different difficulty, the scale and settings of the 3-D feature extractor are variable.

However, in our implementation, a relatively simple and naive network structure is designed to further prove the superiority of our proxy-based learning mechanism in terms

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TABLE I
ARCHITECTURE OF THE SPECTRAL–SPATIAL FEATURE EXTRACTOR

Layer	Kernel Size	Channels	Parameter Number
3D Conv-1	[1, 1, 7]	64	448
3D Conv-2	[3, 3, 3]	16	27,648
3D Conv-3	[3, 3, 3]	32	13,824
3D Conv-4	[3, 3, 1]	32	9,216
flatten	/	/	0
Overall Parameter Number			58,480

of robustness and generalization. Specifically, we choose three sets of 3-D convolution kernels to conduct a four-layer shallow network. With fewer channels, the total parameter volume of the network is less than 60k. Compared to other mainstream methods, the scale of our feature extraction net is two orders of magnitude smaller 1. The network parameter in detail is shown in Table I and visualized in Fig. 2.

With a data patch $\mathbf{F}_m^{n\times n\times c}$ as input, the network outputs a flattened feature vector $f(\mathbf{F}_m;\theta)$, where θ denotes the network parameters.

C. Proxy-Based Deep Learning

Facing hundreds of dense spectral bands from raw HSI data, CNN outperforms conventional methods by automatically learning higher semantic features from the data. Apart from feature extraction, the CNN classification mostly relies on densely connected layers (such as the full connection (FC) layer), which brings a prohibitively large amount of parameter to the architecture. As aforementioned in Section I, this complexity while brings disadvantages on efficiency and robustness. Also, because the constraints of training only appear in the independent sample and the corresponding label, traditional CNN is lack of the ability to optimize the distribution of embedding feature space, e.g., the projected samples of the same class may have lower similarity than ones from other class. These drawbacks of CNN-based HSI classification methods limit the further improvement of HSI-related remote sensing tasks.

In our proxy-based deep learning, the extracted deep feature is classified by the class-represent proxies, which are trained jointly with the CNN feature extractor. Specifically, the class of each projected instance is determined by the class of the closest proxy in the embedding feature space. Every proxy can be viewed as a metric center of same-class instance vector in this feature space, which has the same size with instance feature $f(\mathbf{F}_m;\theta)$. The advantages of using proxy learning are fourfold.

1) Due to the dense back end of CNN is replaced by few-parameter proxies, the scale of the whole network is greatly reduced.
2) With high representative proxies, the demand for complex feature extraction network is also reduced, which brings high computing efficiency to our framework.
3) Because the proxies are projected by a certain type of data, which indicates the parameters naturally approximates the convergence state, the convergence of proxy learning is much faster than other existing CNN.
4) Due to the clear intraclass and interclass decision boundary (achieved with proper loss functions), the robustness to noises and outliers is noticeably improved.

Algorithm 1 Training Procedure of the Proposed Proxy-Based Framework

Require: The input hyperspectral data patch sets $\mathbf{F}_m, m \in 0, 1, \ldots, N_{labeled}$, the network $f(x;\theta)$, early batches including all classes $\mathcal{B}$, empty proxy sets P.

Initialization $p_i = \frac{1}{|\mathcal{B}_i|}\sum_{i\in\mathcal{B}_i} f(\mathbf{F}_m;\theta),\ p_i \in P$

1: **repeat**
2: Obtain the feature $f(\mathbf{F}_m;\theta)$;
3: Compute differences A between $f(\mathbf{F}_m;\theta)$ and P based on distance/similarity based metric;
4: Calculate the $\mathcal{L}_{DisFinal}/\mathcal{L}_{Sim}$ based on A;
5: Update the proxies P and the network θ;
6: **until** Convergence or exceeding max epochs

The target of the proxy-based learning is to optimize the proxies and network to the ideal state for classification, which indicates that the distribution of proxies is a sufficient division of the feature space, and the feature extracted by the network is able to fully express the difference between categories in this feature space. The learning procedure is simply composed of two stages.

1) *Initialization Stage:* Establishment of the origin proxy for each class and initialize the network.
2) *Update Stage:* Refining these proxies and networks jointly by backpropagation of custom proxy losses.

In terms of specifics, the proxy-based deep learning procedure is described in Algorithm 1, whose convergence state is defined as the epoch when the model has the best validation accuracy.

In the training stage, the framework will initialize and update the deep proxies list. The proxies are noted as $p_{i,j} \in P$, in which $i \in 1, 2, \ldots, N_{\text{class}}$ is the index of classes and $j \in 1, 2, \ldots, N_{\text{proxy}}$ is the index of proxies in single class.

For initialization, the proxies are the mean value of the first batch of data D projected into the embedding feature space; if the first batch does not include all category, the proxies are set to zero until the instance of the absent category is fed. It is also acceptable that initialization by zero vector or random vector but efficiency will be affected for the longer converge path.

An important feature of proxy learning is scalability. As the number of proxies of each class increases, the more complex feature embeddings can be covered, conducting a more representative model [46]. Though the proxy number of each category can be varied, while, in our test, one proxy for each category is able to reach state-of-the-art level results in most HSI data sets. Also considering the computing efficiency, in the following part of this article, the number of proxies in each class is set to one, i.e., the number of all proxies is equal to the class numbers. By updating, the proxies can be viewed as the last layer of the network, which is end-to-end optimized altogether with the feature extraction network. To train the

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framework effectively and robustly, comprehensively metrics about the difference between the proxies and instance features need to be considered. Inspiring the works from metric learning, there two major categories of metrics adopted for the loss functions to evaluate the difference between proxies and instance features: the Euclidean distance and cosine similarity.

D. Proxy Loss Functions

Proxy Loss Design Guidelines: To make the whole framework end-to-end trainable, the loss functions need to satisfy two requirements: 1) the loss functions must be derivable to both CNN and proxies and 2) the optimization of the loss functions makes the embedded feature close to the proxy of the corresponding class, meanwhile keeping the distance of the proxies of different classes. To satisfy the abovementioned requirements, while considering two common metrics in metric learning, we construct loss functions of proxy-based HSI classification for both Euclidean distance and cosine similarity.

1) Euclidean Distance Loss Functions: Intuitively, in the embedding space, the instance feature points tend to scatter on a low-dimensional manifold, in which, if one instance feature is spatially closer to one proxy than others, then this instance should be assigned with the class of this proxy.

Following the intuition, the first loss function adapts probabilistic Euclidean distance between each instance feature vector and the closet proxy, which is designed as the distance-based cross-entropy (DisCE) loss.

First, because the Euclidean distance between the proxy and the feature vector of instance is utilized to predict the class, the probability of correct classification $P(\mathbf{F}_m \in C_i)$ is positively correlated with this distance from samples to the correct proxy

$$P(\mathbf{F}_m \in C_i) \propto -\| f(\mathbf{F}_m; \theta) - p_i \|_2^2 \tag{1}$$

where p_i represents the proxy of class C_i. The classical cross-entropy loss (CE) is denoted as $\mathcal{L}_{\text{CE}}(P) = \sum_{i=1}^{m} -y_i \log P$, where m is the total number of training samples, y_i is the one-hot label of training instance x_i, and the probability of sample belongs to the class is indicated by P. Similarly, we normalize the distance-related probability for nonnegative and sum-to-one properties

$$P(\mathbf{F}_m \in C_i) = \frac{\exp\left(-\alpha \| f(\mathbf{F}_m; \theta) - p_i \|_2^2\right)}{\sum_{n=1}^{C} \exp\left(-\alpha \| f(\mathbf{F}_m; \theta) - p_n \|_2^2\right)} \tag{2}$$

where C is the number of classes, i.e., the number of proxies, and α is a hyperparameter that controls the scale of distance constraint. Then, we update the probability function into the final distance-based cross entropy loss (DisCE) as

$$\begin{aligned} \mathcal{L}_{\text{DisCE}} &= -\log(P(\mathbf{F}_m \in C_i)) \\ &= -\log\left(\frac{\exp(-\alpha \| f(\mathbf{F}_m; \theta) - p_i \|_2^2)}{\sum_{n=1}^{C} \exp(-\alpha \| f(\mathbf{F}_m; \theta) - p_n \|_2^2))}\right). \end{aligned} \tag{3}$$

When it comes to multiple-proxy situation, the distance sum calculation of proxies is easily extended to every proxy in each class while the numerator maintaining the estimation of the distance to the closest proxy. It is easy to figure out that with the optimizing of the DisCE loss, the distance from each feature vector to the proxies of its genuine class is decreasing. In our implementation, by adapting the $\mathcal{L}_{\text{DisCE}}$ only, the framework is able to achieve the mainstream deep-learning-based HSI classification performance. However, the implementation with single cross-entropy loss has similar disadvantages appear in conventional CNN, especially the unclear constraints of feature space separation within categories, resulting in relatively low robustness to generalization.

To further improve the performance on the accuracy, we wish that the loss functions can improve the spatial distribution compactness of proxies and corresponding instance feature, which means better discrimination ability. The key point of better spatial distribution is keeping the distance of proxies from different classes as far as possible while maintaining the output feature of the same class close to their center, which is assembled of proxies or one proxy. Thus, we propose the proxy regularization (PR) loss. Following the softplus function, we have:

$$\mathcal{L}_{\text{PR}}(D) = \log(1 + \exp(D)) \tag{4}$$

where D denotes intraclass/interclass distance.

When only intraclass constraint is considered, D is defined as the distance between one output feature and the closest proxy P_i from corresponding class i

$$D_{\text{intra}} = \| f(\mathbf{F}_m; \theta) - p_i \|_2^2. \tag{5}$$

For extremely thorough feature space separation, the constraints between classes (interclass) will also be introduced. Moreover, by adding interclass constraints, margins between categories can be introduced to achieve stronger isolation of decision boundaries for each prototype, as shown in Fig. 2. Considering a decision margin τ, the interclass distance is designed as

$$D_{\text{inter}} = \sum_{j \neq i}^{C} (\tau - \| f(\mathbf{F}_m; \theta) - p_j \|_2^2). \tag{6}$$

Combining the intraclass constraint and interclass constrains to the regularization, a complete total loss considering both interclass/intraclass regularization and margin is design as

$$\mathcal{L}_{\text{DisTotal}} = \mathcal{L}_{\text{DisCE}} + \lambda \mathcal{L}_{\text{PR}}(D_{\text{inter}} + D_{\text{intra}}) \tag{7}$$

where λ is a weight hyperparameter that balances the strength of regularization.

However, in our tests, we have found that, if giving strong, fixed constraints to the embedded feature with proxies of other categories, there will be slight performance degradation in the HSI classification task. In order to explore this issue deeply, multiple combinations for regularization are studied in ablation study in experiments. As the result, reaching for the best performance, the final training loss function is the classification loss ($\mathcal{L}_{\text{DisCE}}$) integrated with intraclass regularization loss by the similar reweighing hyperparameter λ, which is given as follows:

$$\mathcal{L}_{\text{DisFinal}} = \mathcal{L}_{\text{DisCE}} + \lambda \mathcal{L}_{\text{PR}}(D_{\text{intra}}). \tag{8}$$

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2) Cosine Similarity Loss Functions: To further improve the robustness of the proposed method, cosine similarity is introduced as the second metric. Different from the Euclidean distance, in proxy-based learning, cosine similarity concentrates on the angle differences between embedded vectors and proxies, which mitigates the impact of noisy data. More deeply, from the perspective of the Fourier transformation, if decomposing an HSI data cube into magnitude and phase in the frequency domain, then it is not possible to reconstruct a correct cube with a random phase and original magnitude, but it is possible with random magnitude and original phase. Therefore, the phase can better retain the low- and high-level information in the picture, while the magnitude is not essential. When comes to the HSI classification mission, the abovementioned characteristics enable the cosine similarity to focus on the direction than the scale, which brings higher robustness to outlier and noise in very high-dimensional data. Expanding this idea into proxy-based HSI classification, the cosine similarity-based proxy loss function is designed.

The basic idea of similarity-based proxy loss is identical to the distance-based ones, which is increasing the similarity within the class and reduce the similarity between the classes as much as possible. Following this basic idea, the similarity-based proxy loss is initially formulated by the reweighted difference between the instance feature and the intraclass/interclass proxies:

$$\mathcal{L}_{\text{Sim}} = \mathcal{L}_{\text{CE}}(p \cdot S_{\text{intra}} - q \cdot S_{\text{inter}}) \tag{9}$$

where S_{intra} is the sum of similarity between the instance feature and the proxies of the corresponding class, and S_{inter} is the sum of similarity between the instance feature and other proxies. p and q are the self-reweighed parameters inspired by the circle loss [57], which enables the framework to dynamically adjust the strength of optimization according to the different optimization status, achieving more flexible optimization and clearer convergence target. Originally, p and q are defined as

$$\begin{cases} p = [O_{\text{intra}} - S_{\text{intra}}]_+ \\ q = [S_{\text{inter}} - O_{\text{inter}}]_+ \end{cases} \tag{10}$$

where O_{intra} and O_{inter} are the optimum values, and $[\cdot]_+$ indicates the "cutoff at zero," which makes sure that factors p and q are nonnegative. In our implementation, to simplify the hyperparameters, also considering the margin m between different classes, O_{intra} is defined as $1+m$, and O_{inter} is defined as $-m$.

By adapting margin m to similarity computing, the final cosine similarity loss is given by

$$\mathcal{L}_{\text{Sim}} = \mathcal{L}_{\text{CE}}(p(S_{\text{intra}} - (1-m)) - q(S_{\text{inter}} - m)). \tag{11}$$

Compared with distance-based losses, similarity-based loss relatively lacks spatial separability, i.e., the impact of the amplitude information of the HSI instance feature is weakened, resulting in a slight performance decreasing in extreme classification. On the other hand, the high-level coupling relationship between the feature and the prototype has been emphasized, which results in higher robustness and lower training data demanding when $\mathcal{L}_{\text{Sim}}$ is adapted for training.

IV. Experiments

To verify the effectiveness of our proposed proxy-based deep learning framework, multiple comparative experiments are conducted in this section. First, we will introduce the used data sets and the details of the implementation of our method. Then, the results and relative discussion will be reported.

A. Data Sets and Settings

There are four publicly available data sets involved in our experiments. They are Indian Pines, Salinas Scene, University of Pavia, and DFC-2013 Houston.

1) Indian Pines: Indian Pines was captured by the Airborne Visible Infrared Imaging Spectrometer (AVIRIS) sensor in Northwestern Indiana, USA. This data set consists of 145×145 pixels and 224 spectral–spectral bands in the wavelength range 0.4–2.5 μm. The Indian Pines scene mainly contains an agriculture field and some forest or other natural perennial vegetation fields. The ground truth of each pixel is designated into 16 classes and is not all mutually exclusive. By removing bands covering the bands of water absorption, [104–108], [150–163], and 220, the overall number of bands is reduced to 200.

2) Salinas Scene: It is gathered by AVIRIS as well, with the same wavelength range and removed spectral. This data set is collected over Salinas Valley, CA, USA, which has a more sophisticated resolution of 512×217. The Salinas scene also contains 16 land-cover classes for the classification task.

3) University of Pavia: It is gathered by ROSIS sensor over Pavia, Northern Italy. This data set has a size of 610×340 pixels with 103 spectral bands within the wavelength range of 0.43–0.86 μm. The University of Pavia pixels are divided into nine ground-covering classes.

4) DFC-2013 Houston: It [58] is gathered by the Compact Airborne Spectrographic Imager (CASI) over the University of Houston campus and the neighboring urban area in 2012. This data set has a size of 349×1905 pixels with 144 spectral bands within the wavelength range of 0.38–1.05 μm. The labeled pixels in the DFC-2013 Houston are divided into 15 ground-covering classes.

More specifically, details about the first three data sets and training/test volumes are shown in Table II, and details about DFC-2013 Houston are shown independently in Table III.

In data sampling, surely, the performance of classification will be improved for more discrimination information while considering the overall spatial volume of mainstream HSI data sets, e.g., Indian Pines has a resolution at 145×145, and oversized patch, such as 25×25, is "uneconomic" for these relatively small spatial scale data set. Also, large patch size will affect the computing complexity, and the oversized patch is hard for shallow networks to extract semantic information. As a result, in our experiments, a fair 11×11 patch size is used for 2-D/3-D convolutional-based methods and the proposed proxy-based learning framework. For the same reason, the PCA target dimension of HSI preprocessing is set to 30 for all three data sets.

All the tests are carried out on a PC with an Intel 9600KF CPU and a single Nvidia GTX1070 graphics

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TABLE II

DATA SET DETAILS ABOUT PAVIA UNIVERSITY, SALINAS AND INDIANA PINES. THE NUMBERS OF TRAIN SET AND VAL SET ARE EQUAL (THE BACKGROUND CATEGORY IS NOT INCLUDED)

University of Pavia (UP)			Salinas (SA)			Indiana Pines (IP)		
#	Class	Samples	#	Class	Samples	#	Class	Samples
1	Asphalt	6631	1	Brocoli_green_weeds_1	2009	1	Alfalfa	46
2	Meadows	18649	2	Brocoli_green_weeds_2	3726	2	Corn-notill	1428
3	Gravel	2099	3	Fallow	1976	3	Corn-mintill	830
4	Trees	3064	4	Fallow_rough_plow	1394	4	Corn	237
5	Painted metal sheets	1345	5	Fallow_smooth	2678	5	Grass-pasture	483
6	Bare Soil	5029	6	Stubble	3959	6	Grass-trees	730
7	Bitumen	1330	7	Celery	3579	7	Grass-pasture-mowed	28
8	Self-Blocking Bricks	3682	8	Grapes_untrained	11271	8	Hay-windrowed	478
9	Shadows	947	9	Soil_vinyard_develop	6203	9	Oats	20
Overall		42776	10	Corn_senesced_green_weeds	3278	10	Soybean-notill	972
			11	Lettuce_romaine_4wk	1068	11	Soybean-mintill	2455
	Sample Allocation		12	Lettuce_romaine_5wk	1927	12	Soybean-clean	593
Dataset	Train/Val Set	Test Set	13	Lettuce_romaine_6wk	916	13	Wheat	205
UP	6421	29934	14	Lettuce_romaine_7wk	1070	14	Woods	1265
SA	2709	48711	15	Vinyard_untrained	7268	15	Buildings-Trees-Drives	386
IP	1542	7165	16	Vinyard_vertical_trellis	1807	16	Stone-Steel-Towers	93
Overall	10672	85810	Overall		54129	Overall	10249	

TABLE III

DATA SET DETAILS ABOUT DFC-2013 HOUSTON (THE BACKGROUND CATEGORY IS NOT INCLUDED)

#	Class	Train/Val Set	Test Set	Samples
1	Healthy grass	69	1236	1374
2	Stressed grass	73	1308	1454
3	Synthetic grass	40	715	795
4	Trees	63	1138	1264
5	Soil	65	1168	1298
6	Water	17	305	339
7	Residential	74	1328	1476
8	Commercial	68	1218	1354
9	Road	78	1398	1554
10	Highway	71	1282	1424
11	Railway	78	1410	1566
12	Parking Lot 1	71	1287	1429
13	Parking Lot 2	31	570	632
14	Tennis Court	25	463	513
15	Running Track	40	718	798
Overall		863	15544	17270

card. The learning rate of all tests starts from 0.001, and it decays every two epochs by 5%. The optimizer is set to Adam in all tests, with β_1 at 0.9 and β_2 at 0.999. For hyperparameters in the proposed loss functions, α from $\mathcal{L}_{\text{DisCE}}$ is set to 1, β from $\mathcal{L}_{\text{DisTotal}}$ is set to 0.1, and m from $\mathcal{L}_{\text{Sim}}$ is set to 0.1. More specifically, experiments on hyperparameters can be found in Section IV-B3.

For quantitative evaluation, three common evaluation metrics are considered in the following experiments: overall accuracy (OA), average accuracy (AA), and kappa coefficient (kappa). Particularly, OA represents the ratio of correctly classified pixels to test samples in all categories; AA is the average accuracy among each class; and the kappa coefficient is a statistical index representing the agreement level between the ground truth and the actual results, which means that the higher kappa coefficient stands for better classification results. In this article, all kappa results are shown in percentage form.

All tests have been repeated for seven times to record the mean value and standard deviation.

B. Experimental Results and Analysis

1) Quantitative and Qualitative Experiments: In this section, we compare the proposed framework with respect to several most well-known HSI classification approaches, e.g., the kernel SVM [59] and 2-D/3-D convolutional network-based classifier [28]. Meanwhile, several state-of-the-art methods are also introduced into the comparisons, including SSRN [12], HybridSN [11], and S3EResNet [14]. It is worth noted that, in the UP test, we also introduce CapsuleNet [1], DPResNet [13] and convolutional prototype learning (CPL) [46] as comparisons. Besides, to further validate the effectiveness of the proposed four-layer feature extractor, a similar-scale modified ResNet-6 ($\sim$60k trainable parameters) is also introduced to test. The results of all aforementioned approaches are performed locally in our platform, except for CapsuleNet [1], DPResNet [13], and S3EResNet [14], whose results are employed from the original work.

In this series of experiments, all hyperparameters have been fixed to their default setting. In the tests on UP and IP data sets, 15% labeled samples are used for training, another 15% are used for validation. As for the SA and DFC-2013 Houston data sets, 5% labeled samples are used for training and validation, respectively. All the rest of the labeled samples are for tests. First, quantitative performance tests are taken on all three abovementioned HSI data sets. Tables IV–VII present the HSI classification results in OAs, AAs, and kappa coefficients.

We can observe that the proposed method based on the Euclidean distance metric achieves the highest performance among all mentioned methods, whose the value of OA is able to reach 99.99% and average on 99.98% when testing on the UP data set. Meanwhile, the cosine-similarity-based implementation reaches a slightly lower accuracy while still noticeable higher than other competitors. The results of the similar-scale modified ResNet-6 are more than two percentage points lower than the proposed implementations, which further proves the superiority of compactness and performance

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TABLE IV

CLASSIFICATION RESULTS FOR UNIVERSITY OF PAVIA DATA SET USING 15% OF LABELED DATA FOR TRAINING AND 11 × 11 INPUT PATCH. * INDICATES USING 20% OF TRAINING SAMPLES

	OA	AA	Kappa
SVM	93.44±0.98	92.77±1.14	93.01±0.86
2D CNN	95.77±0.89	94.99±1.01	95.45±0.88
3D CNN	99.14±0.19	98.77±0.42	98.97±0.27
CPL [46]	98.13±0.28	97.85±0.33	98.07±0.21
ResNet-6	97.66±0.58	97.12±0.78	97.53±0.89
SSRN [12]	99.86±0.10	99.78±0.12	99.81±0.14
HybridSN[11]	99.76±0.05	99.58±0.12	99.69±0.07
CapsuleNet[1]	99.95±0.02	99.90±0.05	99.93±0.03
DPResNet [13]	99.94±0.01	99.89±0.05	99.32±0.02
S3EResNet* [14]	99.93±0.03	/	/
Ours (Dis)	**99.98**±0.01	**99.95**±0.03	**99.96**±0.02
Ours (Sim)	99.96±0.02	99.94±0.03	99.94±0.02

TABLE V

CLASSIFICATION RESULTS FOR SALINAS DATA SET USING 5% OF LABELED DATA FOR TRAINING AND 11 × 11 INPUT PATCH

	OA	AA	Kappa
SVM	76.24±1.08	77.43±0.79	73.03±1.11
2D CNN	87.25±2.12	86.32±2.72	85.70±2.13
3D CNN	90.30±2.21	90.22±2.94	89.20±1.76
SSRN [12]	99.18±0.10	98.76±0.09	98.86±0.11
HybridSN [11]	98.28±0.31	99.13±0.27	98.08±0.39
S3EResNet [14]	99.14±0.20	99.56±0.10	99.04±0.20
Ours (Dis)	99.79±0.06	**99.85**±0.08	99.76±0.07
Ours (Sim)	**99.85**±0.03	99.84±0.02	**99.83**±0.03

TABLE VI

CLASSIFICATION RESULTS ON INDIAN PINES. * INDICATES USING 10% OF TRAINING SAMPLES (5% LESS THAN OTHERS)

Method	OA	AA	Kappa
SVM	84.37±1.25	82.66±1.56	83.41±1.76
2D CNN	87.81±3.46	84.95±2.55	85.36±1.93
3D CNN	96.72±0.38	95.19±0.95	96.72±0.77
SSRN [12]	98.92±0.27	90.97±2.16	98.72±0.53
HybridSN [11]	97.73±0.16	97.42±0.11	97.42±0.21
S3EResNet* [14]	97.76±0.40	97.45±0.40	96.28±0.40
Ours (Dis)	98.94±0.10	**99.34**±0.07	98.84±0.16
Ours (Sim)	**99.09**±0.12	99.21±0.23	**98.96**±0.22

TABLE VII

CLASSIFICATION RESULTS FOR DFC-2013 HOUSTON DATA SET USING 5% OF LABELED DATA FOR TRAINING AND 11 × 11 INPUT PATCH

	OA	AA	Kappa
SVM	63.68±1.56	77.43±1.88	60.60±2.01
2D CNN	83.34±3.69	85.51±2.83	82.77±3.71
3D CNN	89.30±0.89	91.97±1.04	90.11±0.72
SSRN [12]	96.70±0.91	96.91±0.59	96.43±0.72
HybridSN [11]	92.97±1.52	93.33±1.44	92.40±1.76
Ours (Dis)	98.73±0.30	98.64±0.19	98.69±0.22
Ours (Sim)	**98.82**±0.21	**98.76**±0.18	**98.73**±0.17

of the proposed framework. More significant differences in performance can be found in the tests on lower training ratios, which are demonstrated in Table XII.

In the experiments on the SA data set, both two forms of our implementations reach the highest-level accuracy on a 5% training data ratio. Due to the high robustness of similarity-based loss functions, the similarity-based implementation has an overall higher accuracy in fewer training samples, whose OA is 99.85% on average.

When comes to the IP data set, our methods still have the advantage on the accuracy, while the similarity-based implementation outperforms the distance-based one. In our opinion, this is mainly caused by the nonuniformity of IP data set, including the high diversity on the number of different classes, and mixed class labels, such as the last two categories shown in Table II.

As for the tests on the DFC-2013 data set, the very limited training samples increase the difficulty of classification. Due to the high robustness of the proposed proxy-based framework, the performance advantage to other CNN-based methods is further expanded, such as the similarity-based implementation reaches 98.82% in OA, which is 2.22% higher than the SSRN.

As we mentioned in Section III-D2, the similarity-based feature space distribution is more robust to low-quality input, which is further proved in the next robustness test. Apart from the high accuracy, the results of the proposed framework have the lowest-level of standard deviation, showing well convergence ability. For other deep-learning-based approaches, CapsuleNet reaches the highest performance among them due to the novel capsule architecture. As expected, take HSI data as spectral-only or spatial-only, and conventional methods yield lower accuracy in all three tests.

Corresponding to the quantitative tests, qualitative evaluation results on UP, SA, and IP are shown in Figs. 3–5. Due to the proportion of labeled pixels is rather low, the result visualization of the DFC-2013 Houston data set is omitted.

Overall, the classification maps outputted by the proposed methods are mostly correct and not affected by the over-smoothing issue. Due to the high accuracy of proposed methods, when referring to the ground-truth maps, barely defects can be seen in the visualized results on UP and SA data sets. However, when comes to the IP data set, the 1% error rate results in several noticeable defects in the right part of the image.

It is also worth to be emphasized that our implementation has the lowest scale of parameters of training among all deep-learning-based methods, as we mentioned in Fig. 1. To further prove the efficiency of the proposed framework, the time complexity tests compared with SSRN and HybridSN are presented in Table VIII, and the tests of accuracy verse training time/epochs are presented in Fig. 6. In this series of tests, distance-based losses are utilized in training. Benefit from the efficient deep proxy representation, it can be seen that our proposed method not only demonstrates the lowest time consumption but also achieves the fastest converge speed. In training, the proposed framework can be trained to optimum in dozens of seconds and usually achieves the best performance in less than 20 epochs, which is significantly faster than existing deep-learning-based state-of-the-art methods.

Overall, the proposed proxy-based deep learning framework achieves state-of-the-art level performance on mainstream HSI

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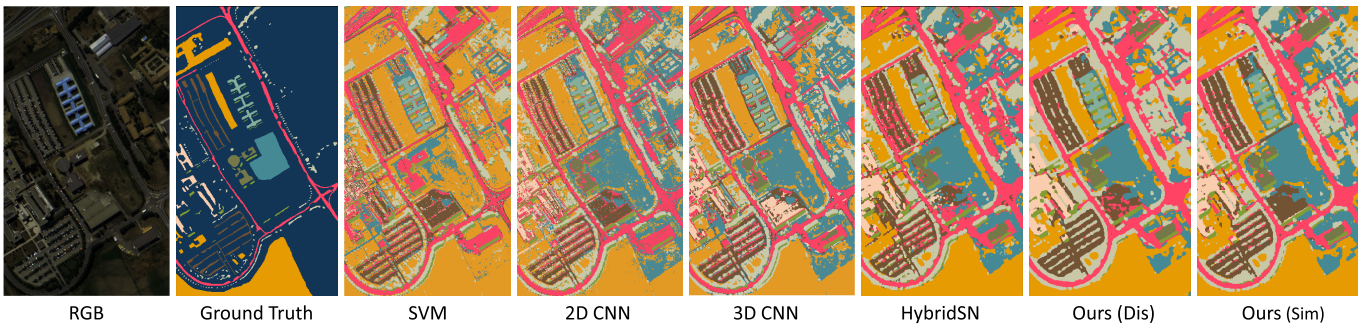


Fig. 3. RGB images, ground-truth maps, and classification maps for the University of Pavia data set.

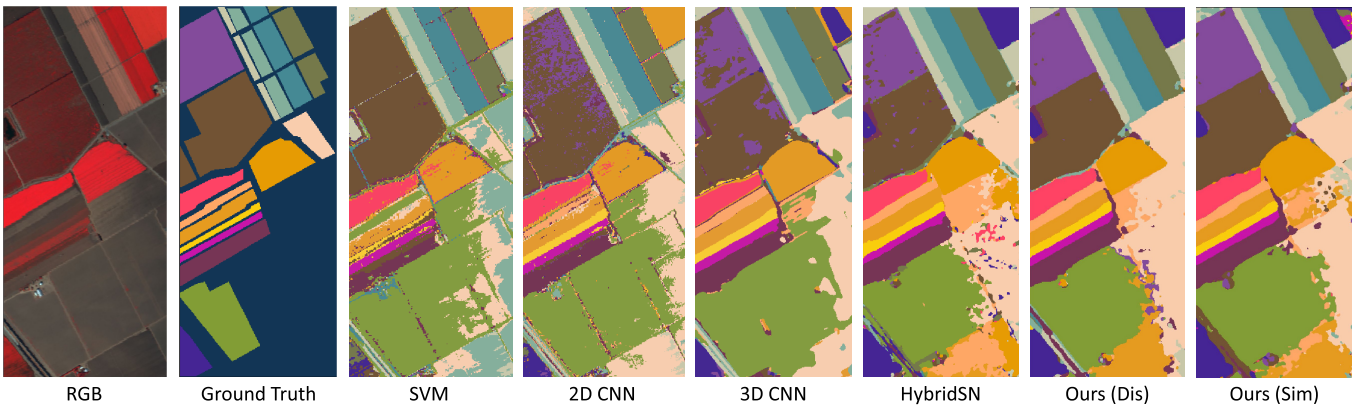


Fig. 4. RGB images, ground-truth maps, and classification maps for the Salinas data set.

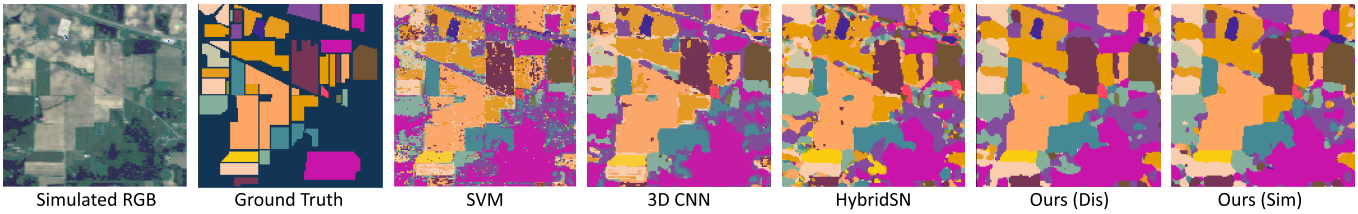


Fig. 5. RGB images, ground-truth maps, and classification maps for the Indian Pines data set.

TABLE VIII

TIME COMPLEXITY TESTS ON THE UNIVERSITY OF PAVIA DATA SET. OURS (FC) IS THE NETWORK INTEGRATED WITH THE PROPOSED FEATURE EXTRACTOR AND THE FC STRUCTURE, WHICH IS OPTIMIZED BY CROSS-ENTROPY LOSS. ALL TESTS ARE PERFORMED WITH A BATCH SIZE OF 32 AND THE DEFAULT LEARNING RATE

Method	Ours	Ours (FC)	HybridSN	SSRN
Time per epoch	**1.33s**	1.38	2.72s	9.32s
Time for testing	**2.12s**	2.47	3.49s	22.53s
Time to best validation	**32s**	71s	297s	2729s

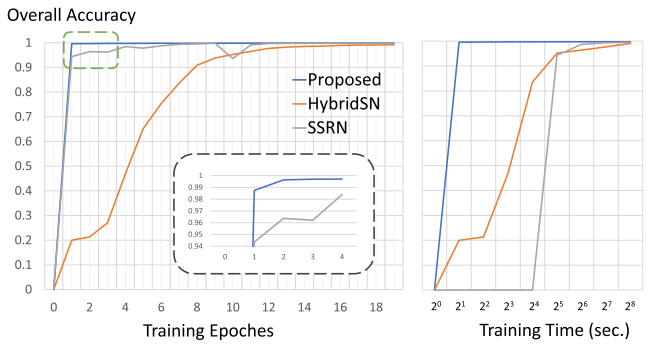


Fig. 6. Overall accuracy verses epochs/training time on the test set of the University of Pavia data set. All methods were trained and tested on the same hardware configuration. Our proxy-based deep learning framework converges significantly faster than other methods while reaching nearly state-of-the-art performance.

data sets while dramatically reducing the requirements on network scale and training time.

2) Robustness Test: In this section, robustness is defined as the performance consistency between the original data and the different-level noised data. To prove the robustness to external disturbances, multiple HSI images noised by different amplitudes are tested with two mainstream methods. To make this experiment more realistic, referring to works focus on HSI noise [60], the Gaussian noise is adapted to affect HSI data as

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TABLE IX

ROBUSTNESS TEST OF THE PROPOSED METHOD, SSRN, AND HYBRIDSN WITH THE UNIVERSITY OF PAVIA DATA SET USING DIFFERENT INTENSITIES OF NOISE AND RATIOS OF TRAINING SAMPLES. STD. IS THE STANDARD DEVIATION OF GAUSSIAN NOISE

		15%			5%		
Std.	Method	OA	AA	Kappa	OA	AA	Kappa
500	SSRN	99.74	99.65	99.65	96.19	94.18	94.95
	HybridSN	98.71	97.64	98.29	92.31	88.47	89.77
	Ours (Dis)	**99.88**	99.77	**99.85**	98.38	97.00	97.85
	Ours (Sim)	**99.88**	**99.82**	99.84	**99.08**	**98.22**	**98.78**
1000	SSRN	99.48	99.46	99.31	92.26	92.50	89.58
	HybridSN	98.02	96.63	97.37	84.42	79.65	79.20
	Ours (Dis)	99.79	99.56	99.72	97.28	95.79	96.37
	Ours (Sim)	**99.83**	**99.60**	**99.77**	**97.85**	**96.86**	**97.13**
2000	SSRN	98.92	98.90	98.57	91.73	88.15	89.12
	HybridSN	96.67	93.99	95.57	79.31	72.58	92.22
	Ours (Dis)	99.62	99.34	99.52	94.63	91.93	92.85
	Ours (Sim)	**99.77**	**99.58**	**99.69**	**97.24**	**95.54**	**96.32**

the white noise. Specifically, on the basis of the original UP data set (max value 8000), we added the Gaussian white noise with the standard deviation of 500, 1000, and 2000. A higher noise standard deviation means a lower signal-to-noise ratio (SNR), which conducts a harder classification. Two levels of training ratios are selected for a comprehensive comparison of robustness, which are 5% and 15%. The patch size of all methods is set to 11×11, 100 epochs are performed on proposed methods, and 200 epochs are performed on SSRN and HybridSN. The results of the robustness test are shown in Table IX.

It can be observed that the performance of HybridSN is noticeable worse in all noise levels. When the standard deviation is 2000, the overall accuracy of HybridSN has decreased significantly, especially in the lower training ratio. Compared to HybridSN, SSRN performs better in all tests with less performance downgradation. Our proposed methods outperform two abovementioned methods in all configuration, i.e., the minimum performance dropping, especially when using the similarity metric, which is 0.08 at 500, 0.13 at 1000, and 0.19 at 2000 on 15% training ratio. When comes to 5% training ratio, the performance gap becomes more significant; at the noises of 2000 standard deviation, the accuracy of the similarity-based implementation is 17.93% higher than the accuracy performed by HybridSN, 5.51% higher than the result of SSRN. The results demonstrate that proxies are still capable to form a relatively accurate representation in the embedded feature space with the growth of the noise and the decrease of training data, which is challenging for conventional CNN networks in noisy training information. The advantage of similarity-based implementation on dealing with noises also proves that the distribution of proxy-based embedding feature is more robust when using a cosine similarity metric.

3) Ablation Study and Hyperparameter Study: In this section, the effectiveness of proposed modules and loss functions is verified through multiple ablation study experiments, and the best value for hyperparameters is also discussed. All tests in this section are performed on the University of Pavia data set with a default 15% ratio of training samples and 11×11 input patch size.

TABLE X

ABLATION STUDY OF THE PROPOSED PROXY-BASED LEARNING FRAMEWORK

Noise Condition	w/o Noise		Scale=2000	
Feature Extractor	Proposed	SSRN	Proposed	SSRN
Proxy (Dis)	**99.98**	99.78	99.62	99.12
Proxy (Sim)	99.96	99.71	**99.77**	99.25
FC+SoftMax	99.85	99.86	98.43	98.92

First, to prove the effectiveness of the proposed proxy-based learning framework, the ablation studies around feature extraction networks and classifiers are performed. In this series of experiments, four different network combinations are tested under two types of noise conditions with both types of proposed loss functions. In the feature extractor part, the feature extraction part of SSRN is integrated as the comparison of the proposed four-layer network. For the classifier, the proposed proxy-based classifier and the conventional FC layer (with SoftMax and cross-entropy loss) are both tested. The specific settings and the results are illustrated in Table X. When coupling with the proposed four-layer network, the proposed proxy-based learning framework demonstrates the best performance in all settings. In addition, our proxy learning also helps the SSRN effectively in dealing with intense noises. On the other side, the performance of the proposed four-layer network is downgraded by integrating a conventional FC classifier, especially in high noise conditions. Overall, the superiorities on accuracy and robustness of the proposed proxy-based learning framework and loss functions are further proved. However, when using the SSRN front end with the proxy-based back end, the performance drops slightly compared with the original SSRN. Besides the settings of SSRN are not optimized for proxy learning, this downgradation is also caused by that the proxy-based classifier encounters difficulties when handling the high semantic-specific feature extracted by the much deeper SSRN in the embedding space with limited dimensions.

Second, a series of tests are conducted to explore the effectiveness of each part of the proposed distance-based loss functions by evaluating the performance of different loss combinations. We test four different combinations of the proposed distance-based proxy loss functions, which are $\mathcal{L}_{\text{DisCE}}$ alone, $\mathcal{L}_{\text{DisCE}}$ with intraclass regularization, $\mathcal{L}_{\text{DisCE}}$ with both intraclass/interclass regularization, and the whole regularization only. The reweight factor of loss λ is set to 0.1 for all regularization. In order to make the difference between the results more significant, the results when using only 5% of the training samples are added to the test.

Table XI shows the comparison results. We can observe that the combination of intraclass regularization outperforms all other combinations with a 99.96% overall accuracy. Meanwhile, we can also observe that the purely using $\mathcal{L}_{\text{DisCE}}$ and distance-based regularization both achieve a lower performance. However, the performance of these two forms

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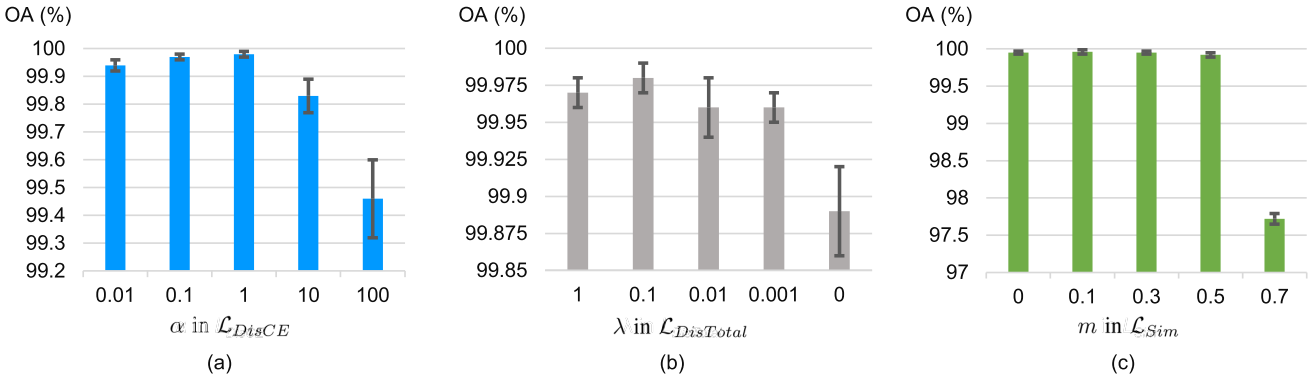


Fig. 7. Impact of different training hyperparameter settings over the performance of the proposed methods. The error bar notes the standard deviation. The subplots (a)–(c) demonstrate the performance on different settings of α in $\mathcal{L}_{\text{DisCE}}$, λ in $\mathcal{L}_{\text{DisTotal}}$., and m in $\mathcal{L}_{\text{sim}}$.

TABLE XI
ABLATION STUDY OF THE PROPOSED DISTANCE-BASED LOSS FUNCTIONS

$\mathcal{L}_{DisCE}$	$\mathcal{L}_{PR}(D_{intra})$	$\mathcal{L}_{PR}(D_{inter})$	OA (15%)	OA (5%)
✓			99.89	99.51
✓	✓		**99.98**	**99.76**
✓	✓	✓	99.92	99.57
	✓	✓	99.86	99.44

is still comparable to other mainstream deep-learning-based HSI classification approaches. When training with only 5% samples, the trend of performance is identical, but the gap is more noticeable. Moreover, the introduction of interclass regularization with margin unexpectedly degrades the performance by 0.04% but still has a better performance than using $\mathcal{L}_{\text{DisCE}}$ or regularization alone. In our opinion, this degradation is mainly caused by the samples of different categories with high embedded similarity. In this scenario, strong constraints, such as interclass distance and category margin, may enhance the influence of these hard samples, which leads to a global performance loss. Another potential reason is the fixed margin in the PR. Due to the varying difficulty of different categories and different training stages, the fixed margin in the implementation may affect the convergence and generalization during the training. More importantly, compared to the similarity-based metric, the distribution of proxies on distance-based implementation is less flexible in feature embedding space (as explained in Section III-D2), which results in lower performance when using fixed margin constraints.

Overall, the proposed loss functions enable our framework to conduct comparable results to nowadays mainstream HSI classification approaches, and the best combination of them can reach state-of-the-art performance with a much smaller network scale.

Third, to prove the effectiveness of different hyperparameter settings of the proposed framework, we test all three hyperparameters in a range of possible values.

The results are shown in Fig. 7; please note that the ordinate axes of the three are not the same. On the whole, the proposed framework is not particularly sensitive in hyperparameter, and performance can be maintained at the state-of-the-art level for a particularly wide hyperparameter range. For α, 1 is the optimal value for $\mathcal{L}_{\text{DisCE}}$, and 0.1 is the best value for λ in $\mathcal{L}_{\text{DisTotal}}$. When λ is set to 0, the network is trained by $\mathcal{L}_{\text{DisCE}}$ alone, and the lowest performance further proves the effectiveness of the proposed PR loss (as we mentioned above). In $\mathcal{L}_{\text{Sim}}$, our implementation keeps a state-of-the-art level performance when m is less than 0.5, but the accuracy drops fast when the value exceeds 0.7. Due to the maximum similarity between different sample is 1, the reasonable margin enables the feature space to be more clearly divided by categories, but the over high margin will push proxies too hard to reach a reasonable distribution, resulting in massive performance degradation.

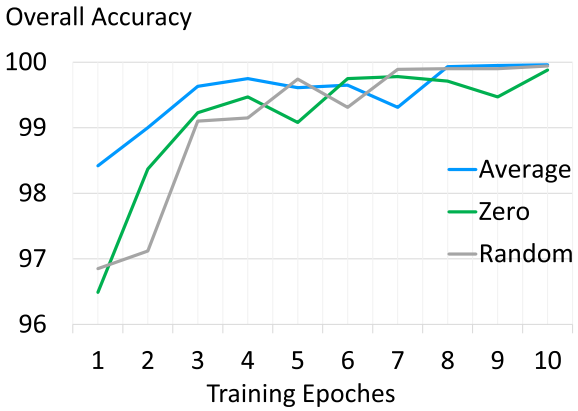


Fig. 8. Impact of different initialization methods on training of the proposed methods.

Finally, to explore the effects of different proxy initialization strategies, the accuracy trends of training by three aforementioned initializations are recorded in Fig. 8. On the whole, the proposed framework is more robust to the selection of initialization methods, and all the implementations will reach the 99.96% or higher accuracy rate in tests. On efficiency and stability, the average initialization outperforms the other two initialization strategies, which can be explained by the shorter convergence path and implicitly distanced interclass embeddings of different class samples. For zero initialization and random initialization, the reasonable embedding distribution

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TABLE XII

IMPACT OF DIFFERENT TRAINING DATA PERCENTAGE OVER THE PERFORMANCE OF SSRN [12], HYBRIDSN [11], S3EBOFRES [14], AND THE PROPOSED METHOD

	SSRN [12]			HybridSN [11]			S3EBoFRes [14]			Ours (Dis)			Ours (Sim)		
Ratio	OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa
15%	99.86	99.78	99.81	99.76	99.58	99.69	/	/	/	**99.98**	**99.96**	**99.95**	99.96	99.94	99.94
10%	99.72	99.47	99.63	99.39	98.93	99.20	99.77	99.69	99.67	**99.94**	**99.93**	99.89	99.93	99.83	**99.91**
5%	99.08	98.78	98.70	98.05	96.95	97.42	99.67	99.56	99.62	99.76	99.61	99.69	**99.83**	**99.75**	**99.77**
3%	98.82	98.38	98.43	96.94	94.87	95.94	99.58	99.36	99.44	99.69	99.42	99.58	**99.74**	**99.58**	**99.65**

can be formed in a few epochs by the guidance of the proposed loss functions, resulting in the approximate performance of mean initialization by the end of the chart.

4) Tests With Different Sample and Patch Sizes: We further explore the performance of the proposed method with different sample and patch sizes. In this series of tests, multiple training ratios and different patch sizes are involved. All tests in this section are performed on the University of Pavia data set.

First, three lower ratios, 3%, 5%, and 10% (default is 15%) of the training samples, are tested on the proposed methods and several SOTA methods. As shown in Table XII, the performance of our proposed framework maintains high accuracy in both metrics and all training data ratios. When testing on 10% training samples, the distance-based implementation keeps the highest OA and AA, but kappa is lower than the more robust similarity-based implementation. As the training ratio comes to 5%, the performance gap between the distance-based implementation and S3EBoFRes shrinks to less than 0.1%, while the similarity-based implementation takes the lead in all evaluation metrics, which is noticeably higher than all other deep-learning-based methods. With the ratio degrades to 3%, the similarity-based implementation shows greater advantages in dealing with less input information, resulting in a 99.74% overall accuracy.

Second, the performance under extremely low training ratios is also tested. In this test, training samples for each class varies from 10 to 100, and the results of the distance-based implementation are shown in Fig. 9. The proposed method reaches 82.15% accuracy when ten samples per class are used for training. When the number of samples per class comes to 30 (0.6% training sample ratio), the accuracy is higher than 96%. These tests in these extreme conditions further prove the robustness of the proposed framework on the low training ratio.

Besides the training sample ratio, we further test the effects on different patch sizes. In this test, limited by the common kernel sizes, the minimal input size is restricted to 7×7. As for the maximum size, considering the computing efficiency and the extraction ability of the four-layer 3-D CNN integrated, the maximum patch size is set to 15×15. Finally, five different patch sizes are evenly defined from 7×7 to 15×15. The training ratios in this test are 3% and 15%. The performances of the proposed methods and the other two CNN-based methods are shown in Table XIII.

It is noted that our framework with distance-based metrics achieves the highest accuracy in all settings. When using

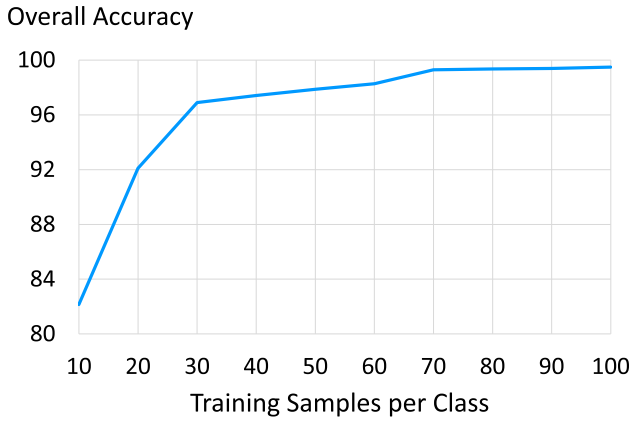


Fig. 9. Results of the distance-based implementation of the proposed framework with different numbers of training samples per class.

TABLE XIII

IMPACT OF DIFFERENT PATCH SIZES OVER THE PERFORMANCE OF THE PROPOSED METHODS AND MULTIPLE MAINSTREAM DEEP LEARNING METHODS. D INDICATES DISTANCE-BASED IMPLEMENTATION, AND S INDICATES SIMILARITY-BASED IMPLEMENTATION

Ratio	Patch	Caps[1]	SSRN[12]	HybridSN[11]	Ours(D)	Ours(S)
15%	7×7	99.75	99.54	/	**99.80**	99.68
	9×9	99.89	99.73	99.17	99.86	**99.95**
	11×11	99.93	99.79	99.76	**99.98**	99.96
	13×13	/	99.87	99.66	**99.97**	99.94
	15×15	**99.97**	99.90	99.82	99.90	99.91
3%	7×7	/	97.94	/	**98.96**	98.61
	9×9	/	99.03	95.57	**99.34**	99.23
	11×11	/	98.82	96.94	99.69	**99.74**
	13×13	/	98.70	97.01	**99.72**	99.67
	15×15	/	99.20	97.46	**99.68**	99.62

smaller patch sizes, the performance of all tested methods is affected by the limited input information, especially in the low training ratio condition. On the other end, when the input spatial size is 15×15, CapsuleNet [1] reaches the best performance (99.97%) among all methods. It is worth noting that our implementations based on both metrics reach the same level of performance with only half input pixels (11×11 versus 15×15). However, unlike other CNN-based methods, the performance of the proposed method does not benefit effectively from larger input patch sizes. The main reason for this phenomenon is the limited feature extracting ability of the integrated small-scale 3-D CNN. Due to the small kernels and shallow depth, it is difficult for this network to extract complete high-level semantics information when facing much larger input. Overall, with the current network scale, the proxy-based framework is robust from low sample sizes and small

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patch sizes. In actual deployment, our proposed framework can adjust the network scale to adapt to the various requirements.

V. Conclusion

In this article, we propose a proxy-based deep learning framework for efficient and robust HSI classification. The highlights of this work include the following.

1) Rather than using densely connected layers to predict the category, our framework utilizes the proxy to represent the embedded class-level feature; by measuring the similarity of sample feature and trained proxies, the classification can be performed accurately with high efficiency and robustness.
2) To adapt this proxy-based training mechanism, we design a series of loss functions with two major metrics: the Euclidean distance and cosine similarity. All these loss functions are capable to compress the distance between the embedded feature and corresponding proxy while implicitly pushing proxies of different categories away from each other.
3) In the evaluation, by integrating a simple four-layer network, the proposed framework outperforms most competing approaches on three HSI data sets by a noticeable range, also achieving the best efficiency and robustness.

In the future, we will explore a more sophisticated proxy mechanism in the HSI classification framework with self-adaptive parameters and proxy numbers. For the data insufficient in HSI-related tasks, we will use the characteristic of fast convergence to achieve accurate classification by fewer training samples. Also, the open-set HSI classification may benefit from the proxy-based framework for the distanced embedding space division. Moreover, we are willing to expand the proxy-based framework to other efficiency-demanding or robustness-demanding remote sensing tasks.

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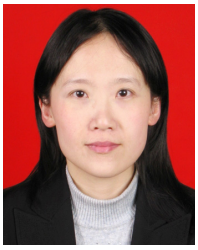
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